

Live Character Robot Software Challenge

Objective

Build a live, expressive character around the supplied fictional 5 degree-of-freedom lamp robot.

For this exercise, the robot uses the laptop camera to see, the laptop microphone to listen, and the laptop speaker for voice, music, and sound effects. The supplied robot model provides the simulated body for motion and light.

The result should not feel like a collection of unrelated AI demonstrations.

Demonstration

Demonstrate one continuous interaction containing these moments:

1. **Engagement:** A person looks toward the character and the character recognizes the opportunity to interact. It disengages appropriately when attention moves elsewhere.
2. **Character response:** It acknowledges the person with an intentional combination of motion, light, voice, sound effects, or music.
3. **Spoken interaction:** It listens through the laptop microphone and responds through the laptop speaker.
4. **Scene memory:** It observes at least one object through the laptop camera, retains useful information about it, and later answers a spoken question about that object.

Across the complete demonstration, show purposeful use of motion, light, voice, a sound effect, and music.

Technical freedom

You may choose:

- Programming languages and frameworks.
- Process and repository structure.
- Simulation or visualization platforms.
- Communication between the character software and simulated robot body.
- Perception, speech, language, and memory approaches.
- Local models, remote services, or a hybrid approach.
- Packaging and deployment method.

You may use cloud APIs, including visual understanding and speech services. If cloud services are used, describe what data is sent, where decisions are made, and why that approach fits the system.

Explain the important choices and their tradeoffs. The supplied URDF defines a robot model, not a required software architecture.

Timebox

Spend no more than **6–8 hours** on the exercise. Prefer a smaller, coherent system with clear decisions over a wide but unfinished implementation. State what you completed and what you intentionally left out.

Supplied files

<https://drive.google.com/file/d/1DUKzg2I3nLi0NnV3QDeb90lSD1G3M3aK/view?usp=sharing>

- [robot/dummy_lamp_6dof.urdf](#)
- [robot/assets/lamp_shade.stl](#)
- [SUBMISSION.md](#)

Use of AI tools

AI-assisted development tools are allowed. You remain responsible for the architecture, behavior, measurements, code, and technical decisions in the submission, and should be able to explain and modify the work as requested/discussed.